



U.S. Department
of Transportation

**Pipeline and Hazardous
Materials Safety
Administration**

April 20, 2023

1200 New Jersey Avenue, SE
Washington, DC 20590

DOT-SP 10326
(FOURTEENTH REVISION)

EXPIRATION DATE: 2027-03-31

(FOR RENEWAL, SEE 49 CFR 107.109)

1. GRANTEE: Honeywell International Inc.
Tempe, AZ
2. PURPOSE AND LIMITATIONS:
 - a. This special permit authorizes the manufacture, mark, sale and use of a non-DOT specification cylinder conforming to all regulations applicable to a DOT specification 3HT cylinder, except as specified herein, for the transportation in commerce of the materials authorized by this special permit. This special permit provides no relief from the Hazardous Materials Regulations (HMR) other than as specifically stated herein. The most recent revision supersedes all previous revisions.
 - b. The safety analyses performed in development of this special permit only considered the hazards and risks associated with transportation in commerce.
 - c. In accordance with 49 CFR 107.107(a), party status may not be granted to a manufacturing permit. These packagings may be used in accordance with 49 CFR 173.22a.
3. REGULATORY SYSTEM AFFECTED: 49 CFR Parts 106, 107 and 171- 180.
4. REGULATIONS FROM WHICH EXEMPTED: 49 CFR § 173.302(a)(1) in that non-DOT specification cylinders are not authorized, except as specified herein.

5. BASIS: This special permit is based on the application of Honeywell International, Inc., dated January 31, 2023, submitted in accordance with § 107.109.
6. HAZARDOUS MATERIALS (49 CFR § 172.101):

Hazardous Materials Description			
Proper Shipping Name	Hazard Class/ Division	Identification Number	Packing Group
Helium, compressed	2.2	UN1046	N/A

7. SAFETY CONTROL MEASURES:

a. PACKAGING: Packaging prescribed is a non-DOT specification cylindrical pressure vessel made from a machined cylinder with an integral machined bottom closure and a circumferentially welded machined top head, having a water capacity not exceeding 260 cubic inches nominal and a maximum service pressure of 10,000 psig. The pressure vessel must be designed and manufactured in conformance with Garret Fluid Systems Division's (GFSD) part 59634-3252938, drawing Nos. 59364-3252938 and 3252938-1 and associated drawings, specifications and other information submitted to and on file with the Office of Hazardous Materials Safety (OHMS). The pressure vessel must also comply with DOT Specification 3HT (§§ 178.35 and 178.44), except as follows:

§ 178.35(e) *Safety devices*

In lieu of the requirements for a safety relief device as prescribed in § 173.302a(a)(2), a heat sensitive pressure relief device (a heat sensitive cutter valve) containing a thermally active powder contained in the sub-assembly as shown in GFSD drawing numbers 3250837, 3257215, 3237537, 3258082-1 and 3258082-2 on file with OHMS are authorized. The cutter valve is an electro explosive device, which is capable of igniting at a temperature not exceeding 500 °F and provides an adequate means for releasing the contents of the cylinder when exposed to a fire situation.

§ 178.44(a) *Type, size and service pressure.*

Cylindrical pressure vessel with integral heads and threaded opening as shown in drawings referenced above. Water capacity not to exceed 260 cubic inches nominal with a maximum service pressure of 10,000 pounds per square inch gauge at 70 °F.

§ 178.44(b) *Authorized material.*

Corrosion and heat resistant Inconel 718 nickel alloy conforming to Aerospace Material Specification (AMS) 5662E. Welding rod must conform to AMS 5832C.

§ 178.44(d) *Manufacture.*

By best appliances and methods; dirt and scale to be removed as necessary to afford proper inspection. Machined cylinder and top head must be welded into the configuration as shown in the drawing Nos. 53964-3252938 and 3252938-1. Welding must be per Military Specification Mil-W 8611A. All weld joints must be 100 percent radiographed and dye-penetrant inspected after heat treatment.

§ 178.44(e) *Welding or brazing.*

Welding as prescribed in § 178.44(d) of this special permit and as described in Honeywell International Inc. drawing #3252938-1 is authorized.

§ 178.44(f) *Wall thickness.*

(1) Same except for substituting 135,000 psi for 105,000 psi in the last sentence.

(2) and (3) * * *

§ 178.44(g) *Heat treatment.*

Each pressure vessel must be heat treated per Mil-H-6875 and GFSD drawings 3252938 and 3252938-1 to obtain the mechanical properties specified in § 178.44(p) of this special permit.

§ 178.44(i) *Hydrostatic test.*

Applies except that the minimum test pressure must be 17,000 psig.

§ 178.44(j) *Cycling test.*

(1) * * *

(2) Cycling testing on a lot basis is not required if three cylinders from the first lot are cycle tested to at least 5,000 cycles with no failure.

(3) In this special permit, a “lot” means a group of cylinders successively produced having the same:

- (i) Size and configuration;
- (ii) Specified material of construction;
- (iii) Process of manufacture and heat treatment;
- (iv) Equipment of manufacture and heat treatment; and
- (v) Conditions of time, temperature and atmosphere during heat treatment.

In no case may the lot size exceed 200 cylinders; however, any cylinder processed for use in the required destructive testing need not be counted as being one of the 200.

(4) * * *

§ 178.44(l) *Flattening test.*

Not required.

§ 178.44(m) *Physical test.*

* * *

(1) To determine yield strength, tensile strength, elongation and reduction in area of material: Required on 2 specimens cut from one pressure vessel out of each lot or from a ring of the same thickness as the production pressure vessel and subjected to the same heat treatment as the finished pressure vessel. The test specimen may be round bar or 1/2 inch wide sheet type and conforms to specimen geometries prescribed in ASTM E-8.

§ 178.44(n) *Magnetic particle inspection.*

Not required. Instead, each pressure vessel must be inspected using apparatus and procedures for liquid penetrant examination in accordance with ASTM E-165-65. Inspection must be performed externally on the finished pressure vessel after hydrostatic test. Evidence of discontinuities, which in the opinion of the independent inspector may appreciably weaken or decrease the durability of the pressure vessel, must be cause for rejection.

§ 178.44(p) *Acceptable results of tests.*

- (1) Not applicable.
- (2) Physical tests.
 - (i) Elongation at least 10% in gauge length not less than 2 inches; reduction in area is 12% minimum.
 - (ii) Ultimate tensile strength is 180,000 psi nominal; yield strength is 150,000 psi nominal.
- (3) Minimum burst pressure is 22,670 psig.
- (4) Cycling tests.

The design must be qualified by cyclic tests prescribed in § 178.44(j) on three completed pressure vessels. Each pressure vessel must be cycled from zero to service pressure at a rate not exceeding 10 cycles per minute, and for a minimum of 5,000 cycles. The pressure vessel may not show any evidence of distortion or failure.

§ 178.44(q) *Rejected pressure vessels.*

- (1) Reheat treatment is authorized; subsequent thereto, acceptable pressure vessels must pass all prescribed tests.
- (2) Repair of welds is authorized after which reheat treatment is required and subsequent thereto, acceptable pressure vessels must pass all prescribed tests.

§ 178.44(r) *Marking.*

- (1) Applies, except that instead of DOT-3HT, pressure vessels must be marked "DOT-SP 10326" followed by the service pressure. Marking by low stress type method such as electro-chemical etching, vibro-pen or laser marking, which does not decrease the integrity of the pressure vessel, is authorized.
- (2) Stamping of elastic expansion is not required.
- (3) Name plates - Not required.

§ 178.44(s) *Inspector's report.*

Inspector's report must be appropriately modified to reflect identification and conformance with this special permit.

b. TESTING: Each pressure vessel may not be filled and shipped unless it has been reinspected and hydrostatically retested every 5 years in accordance with § 180.209(a) as prescribed for DOT-3HT cylinders, except that the rejection elastic expansion criteria does not apply.

8. SPECIAL PROVISIONS:

a. In accordance with the provisions of Paragraph (b) of § 173.22a, persons may use the packaging authorized by this special permit for the transportation of the hazardous materials specified in paragraph 6, only in conformance with the terms of this special permit.

b. A person who is not a holder of this special permit, but receives a package covered by this special permit, may reoffer it for transportation provided no modification or change is made to the package and it is offered for transportation in conformance with this special permit and the HMR.

c. A current copy of this special permit must be maintained at each facility where the package is offered or reoffered for transportation.

d. Each package manufactured under the authority of this special permit must be marked with a registration symbol designated by the Office of Hazardous Materials Safety for a specific manufacturing facility.

e. A current copy of this special permit must be maintained at each facility where the package is manufactured under this special permit. It must be made available to a DOT representative upon request.

f. The cylindrical pressure vessels authorized under this special permit may be used in military weapons systems or military aircraft only. A copy of the inspector's report on the first lot required in § 178.44(s) must be submitted to the OHMS prior to initial shipment.

g. Pressure vessels must be shipped in strong outside packagings in conformance with § 173.301(a)(9).

h. The pressure vessel service life may not exceed 10 years or 10 refills whichever comes first.

9. MODES OF TRANSPORTATION AUTHORIZED: Motor vehicle and cargo-only aircraft.
10. MODAL REQUIREMENTS: A current copy of this special permit must be carried aboard each aircraft used to transport packages covered by this special permit. The shipper must furnish a current copy of this special permit to the air carrier before or at the time the shipment is tendered.
11. COMPLIANCE: Failure by a person to comply with any of the following may result in suspension or revocation of this special permit and penalties prescribed by the Federal hazardous materials transportation law, 49 U.S.C. 5101 et seq:
 - o All terms and conditions prescribed in this special permit and the Hazardous Materials Regulations, Parts 171-180.
 - o Persons operating under the terms of this special permit must comply with the security plan requirement in Subpart I of Part 172 of the HMR, when applicable.
 - o Registration required by § 107.601 et seq., when applicable.

Each “Hazmat employee”, as defined in § 171.8, who performs a function subject to this special permit must receive training on the requirements and conditions of this special permit in addition to the training required by §§ 172.700 through 172.704.

No person may use or apply this special permit, including display of its number, when the special permit has expired or is otherwise no longer in effect.

Under Title VII of the Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (SAFETEA-LU)—“The Hazardous Materials Safety and Security Reauthorization Act of 2005” (Pub. L. 109-59), 119 Stat. 1144 (August 10, 2005), amended the Federal hazardous materials transportation law by changing the term “exemption” to “special permit” and authorizes a special permit to be granted up to two years for new special permits and up to four years for renewals.

12. REPORTING REQUIREMENTS: Shipments or operations conducted under this special permit are subject to the Hazardous Materials Incident Reporting requirements specified in 49 CFR §§ 171.15 - Immediate notice of certain hazardous materials incidents, and

171.16 - Detailed hazardous materials incident reports. In addition, the grantee(s) of this special permit must notify the Associate Administrator for Hazardous Materials Safety, in writing, of any incident involving a package, shipment or operation conducted under terms of this special permit.

Issued in Washington, D.C.:

A handwritten signature in blue ink, appearing to read "W. Schoonover", is written over a faint, light blue circular stamp.

for William Schoonover
Associate Administrator for Hazardous Materials Safety

Address all inquiries to: Associate Administrator for Hazardous Materials Safety, Pipeline and Hazardous Materials Safety Administration, U.S. Department of Transportation, East Building PHH-13, 1200 New Jersey Avenue, Southeast, Washington, D.C. 20590.

Copies of this special permit may be obtained by accessing the Hazardous Materials Safety Homepage at <https://www.phmsa.dot.gov/approvals-and-permits/hazmat/special-permits-search>. Photo reproductions and legible reductions of this special permit are permitted. Any alteration of this special permit is prohibited.

PO: ae